

Vandana Chidanand Mansur

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EDUCATION

University of California | GPA: 4.0/4.0

Davis, California

Master of Science, Electrical and Computer Engineering (ML Track)

09/2025 - 12/2026

Coursework: Unsupervised Learning, Big Data & High-Performance Statistical Computing, Practical AI

Vellore Institute of Technology | GPA: 8.85/10.0

Vellore, India

Bachelor of Technology, Electronics & Communication Engineering

07/2018 - 05/2022

TECHNICAL SKILLS

Languages: Python, Java, SQL **Libraries & Frameworks:** Spring Boot, Maven, React, Node.js, PyTorch, RAG, LangChain, LangGraph

Technologies: RESTful APIs, Microservices, Machine Learning, NLP, Large Language Models (LLMs), Agentic AI, Prompt Engineering, Cloud Computing, Optimization, A/B Testing, Agile/Scrum,

Tools: Git, GitHub, Jupyter Notebook, Google Colab, Docker, Kubernetes, REST APIs, Postman, Datadog, Splunk, AWS, Chroma vector DB, CI/CD, Service Now, Jira, Gen AI, AI Fluency

WORK EXPERIENCE

University of California, ASEE Lab

Davis, California

Graduate Student Researcher

11/2025 – Present

- Built **AEGIS**, an **agentic GraphRAG** framework for hardware vulnerability detection across **9 OpenTitan IPs**, achieving **0.88 single-run** detection probability via a dual-layer knowledge graph and Synopsys VCS pipeline; demonstrated RAG outperforms frontier long-context models at **135x lower cost** across **13** confirmed vulnerabilities.

Optum, UnitedHealth Group

Hyderabad, India

Full Stack Engineer

11/2024 – 09/2025

- Built and deployed a clinical question-answering assistant using a **RAG architecture (OpenAI GPT-4o mini, ChromaDB, OpenAI text-embedding-3-small)**, implementing backend retrieval workflows integrated with **LLM prompts** to deliver accurate, context-aware medical query responses.
- Developed and maintained modern, **scalable web interfaces** (React, Next.js) for member portals serving **50M+** users across **4 business** lines. Optimized member coverage and benefits journeys to **enhance user engagement, informed decision-making**, and accessibility.
- Applied **low-level** design principles to migrate **legacy SOAP APIs to GraphQL**, optimizing request/response logic, payload handling, and resolver architecture, boosting **API performance** by **30%** and advancing scalable, efficient data access for core applications.
- Automated deployment pipelines and led seamless migration from **on-premises to AWS** (EKS, ASM, S3, IAM, Terraform, CloudFront), ensuring robust, **high-availability releases and improved scalability**.

Optum, UnitedHealth Group

Hyderabad, India

Software Engineer

06/2022 – 11/2024

- Delivered secure **REST APIs** in **Java, Springboot** and **Maven** with HIPAA compliance, reducing **database call latency** by **20%**. Enhanced **digital health** platform by integrating EMRs using **bi-directional APIs integrating to Salesforce app**, automated onboarding, visit summaries, care team-member communication, and streamlining clinician workflows for multi-market Senior Community Care expansion.
- **Architected DevOps Infrastructure** for new applications using **Docker** and **GitHub actions**, implementing **automated CI/CD pipelines** that **reduced build cycles by 60%** across **35** projects and **QA process turnaround time by 70%**. Improved **code quality coverage to 90%**. Enabled **zero downtime and high availability**, managed API gateway, created **unified Grafana, Datadog dashboards** to monitor performance, achieved **100% vulnerability resolution on Prisma Compute**, and **saved 30% on cloud operations storage costs**.

PROJECTS

CommunityCare

2026

- Built an accessibility-first AI intake platform using **React, Node.js, WebSockets, Gemini Live API, BackBoard RAG, and MongoDB** to convert voice/text/ASL-style conversations into structured staff dashboard cards with urgency alerts, local resource matches, multilingual support, appointment coordination, and safety guardrails for healthcare, housing, food aid, and support services.

Mémoire de Parfum

2026

- Engineered Arome, a full-stack **AI chatbot** that translates user memories into personalized luxury fragrance blueprints using a hybrid **GraphRAG + LLM** pipeline built on **Neo4j, Weaviate, FastAPI, and Redis**. Designed brand-agnostic architecture targeting the **€7.5B** exception fragrance market, with inline safety guardrails ensuring no medical or therapeutic claims reach consumer-facing output.

Resume Classification with NLP and Transformers

2025

- Built an automated pipeline to classify 2,484 resumes into 24 job categories using cleaning, **lemmatization, TF-IDF/n-grams**, and **DistilBERT embeddings**, reaching **92% accuracy** and **strong macro-F1** performance. Fine-tuned a **66M-parameter DistilBERT** sequence classifier (6-layer, 12-head Transformer) with **AdamW**, learning-rate scheduling, and regularization, leveraging self-attention to distinguish semantically similar roles and **outperform TF-IDF** baselines by **17** accuracy points.

Medical Chain-Deep Learning & Ethereum Blockchain Integrated Secure Electronic Health Record System for COVID-19

2022

- Implemented **UNet** for lung segmentation (CXRs), **InceptionV3** model with an accuracy of **92-95%**. Introduced the **OCR-BERN** system for patients' utility. Used **IPFS** to provide a scalable, secure solution for storing CXR images, Metamask wallet, React.js, **Solidity** and **Flask** API to integrate the Dapp ([demo](#)).